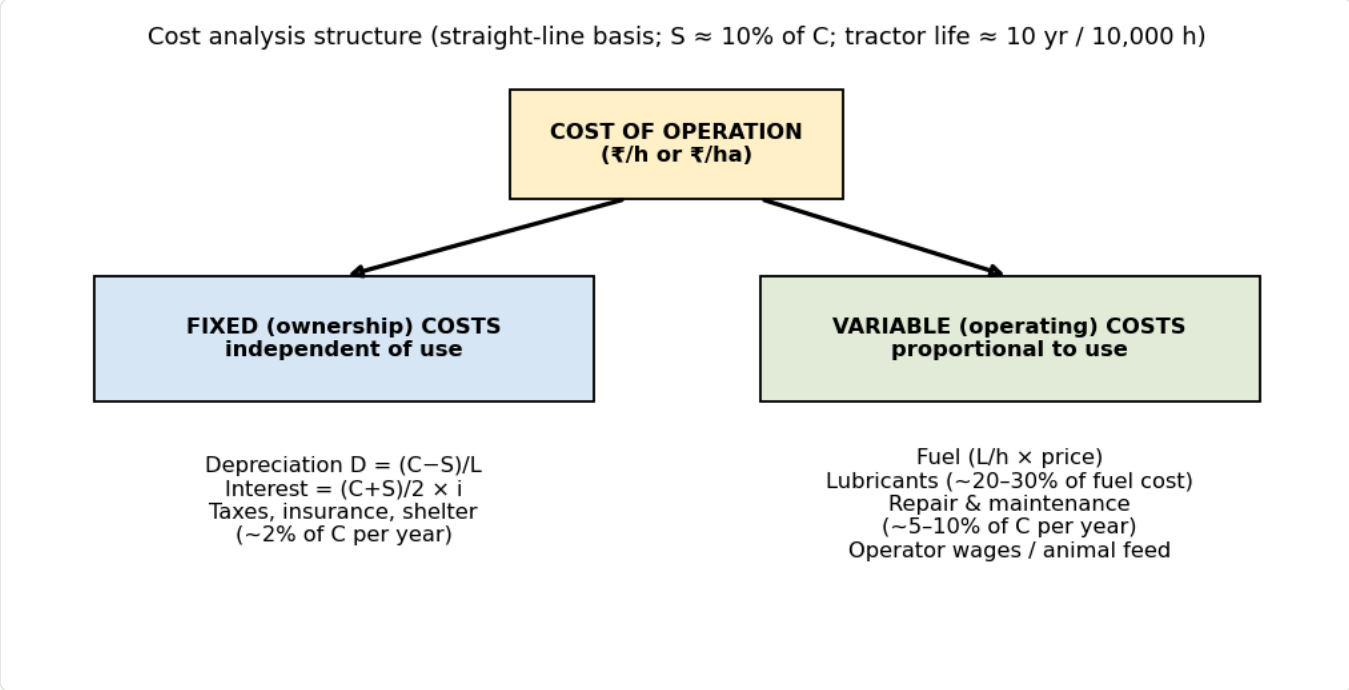
UNIT 3 · TOPIC 9(A)

# Fixed and Variable Costs

*Syllabus topic: Cost analysis of animal and tractor operated implements and tractors*




*Cost analysis structure*

## Fixed (ownership) costs — accrue whether the machine is used or not

$$\text{Straight-line depreciation } D = (C - S)/L \text{ per year}$$

C = purchase price; S = salvage ≈ **10% of C**; L = life in years (tractors **~10 yr/10,000 h**; implements **~2,000–2,500 h**).

| Depreciation method | How it works |
|---|---|
| Straight-line | (C - S)/L each year |
| Declining balance | A fixed % of the remaining value each year — faster early write-off |
| Sum-of-the-years-digits | Fraction (remaining life ÷ Σdigits) × (C - S) — also front-loaded |
| Estimated-value | Actual market value — used for studies |

$$\text{Interest on average investment} = [(C + S)/2] \times i$$

- **Taxes, insurance and shelter (TIS)** ≈ **2%** of purchase price per year (India-typical).
- Fixed cost/h = annual fixed cost ÷ annual hours of use.

## Variable (operating) costs — proportional to use

| Item | Estimate |
|---|---|
| Fuel | Measured L/h × price |
| Lubricants/oil | ≈ 20–30% of fuel cost |
| Repair & maintenance | ≈ 5–10% of purchase price per year; rises with age (over full life, R&M totals roughly 80–120% of C for tractors) |
| Operator | Wages |

## Animal-drawn equipment

Charge the bullock-pair maintenance (feed, care) per working day, plus the implement costs. **Animal cost continues on idle days** — a key economics point.

## Practice MCQs

**Q106. Straight-line annual depreciation is:**

(a) C/L
(b) (C-S)/L
(c) (C+S)/L
(d) C×S/L

**Q107. Salvage value of farm machines is commonly assumed as:**

(a) 1%
(b) 10%
(c) 25%
(d) 50% of purchase price

**Q108. The economic life of an agricultural tractor is usually taken as:**

(a) 2,000 h
(b) 5,000 h
(c) 10,000 h
(d) 25,000 h

**Q109. Interest in cost analysis is charged on:**

(a) Purchase price
(b) Salvage value
(c) Average investment (C+S)/2
(d) Fuel cost

**Q110. Which is a FIXED cost?**

(a) Fuel
(b) Depreciation
(c) Repairs
(d) Wages

**Q111. Which is a VARIABLE cost?**

(a) Insurance
(b) Shelter
(c) Fuel and lubricants
(d) Interest

**Q112. Taxes, insurance and shelter together are typically about:**

(a) 0.2%
(b) 2%
(c) 8%
(d) 15% of purchase price per year

**Q113. Lubricant cost is commonly estimated as:**

(a) 5%
(b) 20–30%
(c) 50%
(d) 75% of fuel cost

**Q120. In sum-of-the-years-digits depreciation, early-year depreciation is:**

(a) Lower than straight line
(b) Higher than straight line
(c) Zero
(d) Equal

**Q121. A key economic feature of bullock power is:**

(a) No fixed cost
(b) Feed cost continues even on idle days
(c) No feed needed
(d) Infinite life

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**Answers**

Q106 — (b) (C-S)/L
Q107 — (b) 10%
Q108 — (c) 10,000 h
Q109 — (c) Average investment (C+S)/2
Q110 — (b) Depreciation
Q111 — (c) Fuel and lubricants
Q112 — (b) 2%
Q113 — (b) 20–30%
Q120 — (b) Higher than straight line
Q121 — (b) Feed cost continues even on idle days

**Sources:** Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 2 (Field Capacities and Cost Analysis); Hunt — Farm Power and Machinery Management; Jagdishwar Sahay; SMAM/CHC guidelines; NPTEL lectures.